



NEXT WAVE (INDIA)

RAMPUR GHAT ROAD, PAONTA SAHIB, DISTT. SIRMOUR (H.P)

REPORT FOR MLT VALIDATION

NAME OF PRODUCT		Effective Date	04/12/2024
Sitagliptin Tablet USP 50 mg			
Report No.	QCM/MVPR/028	Page	1 of 5

REPORT FOR MLT (MICROBIAL LIMIT TEST) VALIDATION OF SITAGLIPTIN TABLET USP 50 MG

SOP No: QCM-019

Format No- QMF/19/03

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Annexure-1

INOCULUM PREPARATION & ESTIMATION OF INITIAL POPULATION

Bacteria		Fungus	
Date of Analysis	: 17/12/24	Date of Analysis	: 05/12/24
Date of Completion	: 20/12/24	Date of Completion	: 10/12/24
Medium Used: SCDA Medium for Bacteria & Media lot No.		:	0000563787
Medium Used: SCA Medium for Fungus & Media load No.		:	0000491278
Incubation Temp. & Time:	30-35°C for 3 days for Bacteria	Incubator ID	: NWI/QCM-14
	20-25°C for 5 days for Fungus	Incubator ID	: NWI/QCM-02

Culture Used	Code No.	Selected dilution	Observation (In cfu)			Growth Observed By/on
			Plate 1	Plate 2	Average	
<i>Staphylococcus aureus</i>	NCIM 2079	10^{-6}	70	74	72	Sm 20/12/24
<i>Pseudomonas aeruginosa</i>	NCIM 2200	10^{-5}	57	61	59	Sm 20/12/24
<i>Escherichia coli</i>	NCIM 2065	10^{-7}	74	88	81	Sm 20/12/24
<i>Candida albicans</i>	NCIM 3471	10^{-5}	64	60	62	Sm 10/12/24
<i>Aspergillus brasiliensis</i>	NCIM 1196	10^{-4}	21	25	23	Sm 10/12/24

Remarks: Inoculum Suspension Count Satisfactory for further validation study.

Checked by: -

Sign/date: -

20/12/24

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RECOVERY OF TESTED MICROORGANISMS IN PRODUCT

Name of Product	:	sitagliptin tablet			Date of Analysis	:	20/12/24	
	:				Date of Completion	:	25/12/24	
B. No.	:	NW035			Method Used	:	pour plate method	
Medium Used: SCDA Medium for TBC & Media Autoclave load No.						:	0000563727	
Medium Used: SDA Medium for TFC & Media Autoclave load No.						:	0000611659	
Incubation Temp. & Time:	:	TBC: 5 days for Bacteria at 30-35°C			Incubator ID	:	NW1/9cm-14	
	:	TFC: 5 days for fungus at 20-25°C			Incubator ID	:	NW1/20cm-02	
Use of Test Organisms	Test Organisms	Dilution	Positive control Plate count (IC)	Product control Plate count (PC)	Product Positive control Plate count (PPC)	Recovery of Tested Microorganisms	Growth Observed By/on	
						Recovery (%) (PPC)-(PC) X 100 IC		
For TBC	<i>Pseudomonas aeruginosa</i>	1:10	54	<10	48	88.89%	MF 25/12/24	
	<i>Staphylococcus aureus</i>	1:10	72	<10	68	94.44%	MF 25/12/24	
	<i>Escherichia coli</i>	1:10	74	<10	66	89.19%	MF 25/12/24	
For TFC	<i>Candida albicans</i>	1:10	54	<10	50	92.59%	MF 25/12/24	
	<i>Aspergillus brasiliensis</i>	1:10	23	<10	21	91.30%	MF 25/12/24	

Negative Control: - No Growth observed

Remarks: Recovery of microorganism in Product Found within Acceptance Criteria.

Checked by: -
Sign/date: 25/12/24

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TEST FOR SPECIFIED MICRO ORGANISM: Escherichia Coli

Name of Product	:	Sitagliptin tablet	Date of Analysis	:	20/12/24
B. No.	:	NW035	Date of Completion	:	27/12/24
Incubation Temperature	:	30-35°C 42-44°C	Incubator ID	:	NWZ/SCM-014 NWZ/SCM-01

S.No.	Name of Materials	Lot No.
1.	Soybean Casein Digest Medium	0000 657607
2.	MacConkey Broth	0000 571727
3.	MacConkey Agar	0000 518050
4.	Gram's stain kit	0000 578162

Observation

S.No.	Test Microorganism	Negative control	Positive control	Product control	Positive Product control
1.	E.coli	No Growth observed	Growth observed	No Growth observed	Growth observed

Remarks:

Analysed By	:	114	Checked By	:	27/12/24
Date	:	27/12/24	Date	:	27/12/24

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4.1 Summary and Conclusion:

Summary: The Product does not exhibit any antimicrobial activity as the recovery of microorganism inoculated observed to be within the acceptance criteria.

a. For TAMC (Total Aerobic Microbial Count) & TYMC (Total Yeast & Mold Count)

Name of the Organism	Acceptance criteria (factor of 2)		Percentage of Recovery of tested organism	Complies /does not comply
	B/2	BX2		
Escherichia coli	40.5	162	89.19 %	Complies
Staphylococcus aureus	36	144	94.44 %	Complies
Pseudomonas aeruginosa	29.5	118	92.31 %	Complies
Candida albicans	31	124	92.59 %	Complies
Aspergillus brasiliensis	11.5	46	91.30 %	Complies

b. For specified Microorganism:

Name of Organism	Growth observed/Not observed	Complies /Does not complies
E.coli	Growth observed	Complies

Conclusion: Based on the above summary it is concluded that the method is suitable for recovery of viable microorganisms and validate the technique of microbial limit test to estimate bioburden level and specified organism recovery. Found Satisfactory.

Recommendation: Further it is recommended that method for sample preparation for sitagliptin shall be "10 gm of sample dissolve in 90 ml of Bubbled sodium chloride peptone solution pH 7.0 with (1% Tween-80) and mix well."

Summarized by (Sign & Date):

HP
27/12/24

Reviewed by (Sign & date):

[Signature]
27/12/24

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